Problems for Time Series Analysis for Financial Data

Quantitative Finance Practice Problems
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Sylvester Cleveland - MSc. Applied and Computational Mathematics

Problems solutions

7.1 Evaluate EZ_t^4 for ARCH(1) process (7.2.5) with $0 < \alpha_1 < 1$ and $\{e_t\} \sim$ IID N(0, 1). Deduce that $EX_t^4 < \infty \iff 3\alpha_1^2 < 1$.

7.1 Sol^n:
Let $m_2 = Z_t^2$, since $Z_t^2 = \alpha_0 e_t^2 + \alpha_1 Z_{t-1}^2 e_t^2$. Taking unconditional expectations for $m_2 = m_{2-1}$ because we assume stationarity, then

$$\begin{aligned} m_2 &= \alpha_0 e_t^2 + \alpha_1 m_{2-1} e_t^2 \\ E[m_2] &= \alpha_0 E[e_t^2] + \alpha_1 m_{2-1} E[e_t^2] \\ m_2 - \alpha_1 m_{2-1} &= \alpha_0 + \alpha_1 m_{2-1} - \alpha_1 m_{2-1} \\ m_2(1-\alpha_1) &= \alpha_0 \\ m_2 &= \alpha_0/(1-\alpha_1) \end{aligned}$$

To take the unconditional expectaions for $E[m_4]$ we have $p = 4$ up to order 4 where

$$h_t = \left(\alpha_0 + \sum_{i=1}^{p=4} \alpha_i X_{t-i}^2\right)^2,$$

with $\alpha_0 > 0$ and $\alpha_j \geq 0$, $j = 1, \ldots, p$. Since we know $X_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$, we compute the conditional expectations as follows

$$\begin{aligned} E[X_t^4] &= E[(\alpha_0 + \alpha_1 X_{t-1}^2)^2] \\ &= 3E[\alpha_0^2 + 2\alpha_0\alpha_1^2 X_{t-1}^2 + \alpha_1^2 X_{t-1}^4], \end{aligned}$$

since $m_2 = \alpha_0/(1-\alpha_1)$ & $m_4 = X_t^4$ then

$$\begin{aligned} m_4 &= 3(\alpha_0^2 + 2\alpha_0\alpha_1^2 m_{2-1}^2 + \alpha_1^2 m_{4-1}^4) \\ &= 3(\alpha_0^2 + 2\alpha_0\alpha_1^2 m_2^2 + \alpha_1^2 m_4^4) \\ m_4(1-\alpha_1) &= 3\alpha_0^2(1-\alpha_1) + 6\alpha_0^2\alpha_1^2 \\ &\quad + (1-\alpha_1)3\alpha_1^2 m_4^4, \\ &\quad \textit{I'll work out the algebra later} \\ &\vdots \\ m_4 &= 3\alpha_0^2(1+\alpha_1)/[(1-3\alpha_1^2)(1-\alpha_1^2)] \end{aligned}$$

The fourth moment converges i.e. is finite *iff* the denomiator is non-zero and positive, which requires $1 - 3\alpha_1 > 0$ since $0 < \alpha_1 < 1$ implies $1 - \alpha_1 > 0$ and all coefficients are positive).

Thus, $E[X_t^4] < \infty$ (converges) iff $3\alpha_1^2 < 1$. ■

$$E[Z_t^4] = \frac{3\boldsymbol{\alpha_0} \cdot (1+\alpha_1)}{(1-3\alpha_1^2)(1-\alpha_1^2)}$$

□

7.2 Let $\{Z_t\}$ be a causal stationary solution of the ARCH(p) equations (7.2.1) and (7.2.2) with $EZ_t^4 < \infty$. Assuming that such a process exists, show that $Y_t = Z_t^2/\alpha_0$ satisfies the equations

$$Y_t = e_t^2 \left(1 + \sum_{i=1}^{p} \alpha_i Y_{t-i}\right)$$

and deduce that $\{Y_t\}$ has the same autocorrelation function as the AR(p) process

$$W_t = \sum_{i=1}^{p} \alpha_i W_{t-i} + e_t, \quad \{e_t\} \sim \text{WN}(0,\ 1).$$

(In the case $p = 1$, a necessary and sufficient condition for existence of a causal stationary solution of (7.2.1) and (7.2.2) with $EZ_t^4 < \infty$ is $3\alpha_1^2 < 1$, as shown by the results of Section 7.2 and Problem 7.1.)

7.2 Sol^n:
The ARCH(1) process with recurssions $Z_t = \sqrt{h_t} e_t$, $\{e_t\} \sim$ IID N(0, 1), where h_t is the positive violatility of $\{Z_s, s < t\}$, defind as

$$h_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Z_{i-i}^2,$$

with $\alpha_0 > 0$ & $\alpha_j \geq 0$, $j = 1, \ldots, p$. Since we've determined that for an $m_2 = E[Z_t^2]$, then

$$Z_t^2 = \alpha_0 e_t^2 + \alpha_1 Z_{i-1}^2 e_t^2$$

Letting $Y_t = Z_t^2/\alpha_0$ we see that

$$
\begin{aligned}
Y_t &= e_t^2\left(1+\sum_{i=1}^{p}\alpha_i Y_{t-i}\right)\\
Z_t^2/\alpha_0 &= e_t^2\left(1+\sum_{i=1}^{p}\alpha_i Z_{t-i}^2/\alpha_0\right)\\
Z_t^2 &= e_t^2\left(\alpha_0+\sum_{i=1}^{p}\alpha_i Z_{t-i}^2\right)
\end{aligned}
$$

because the solution is causal and we're assuming stationarity.

The last term has expectation $\alpha^{n+1}EZ_t^2$ and converges to zero as $n\to\infty$. So, therefore

$$
\begin{aligned}
m_2 &= \alpha_0\left|\sum_{j=0}^{\infty}\right|_0^{\text{1 Great!}}\alpha_1^j e_t^2 e_{t-1}^2\dots e_{t-j}^2\\
m_2 &= \alpha_0 E[e_t^2]+\alpha_1 m_{2-1}E[e_{t-1}^2]\\
m_2-\alpha_1 m_2 &= \alpha_0+\alpha_1 m_2-\alpha_1 m_2\\
m_2(1-\alpha_1) &= \alpha_0\\
m_2 &= \alpha_0/(1-\alpha_1)
\end{aligned}
$$

As was shown in the previous solution when $3\alpha_1^2<1$ then $EZ_t^4<\infty$ i.e. converges. Hence this means that $Y=Z_t^2$ has the same ACF as the AR(1) process as $W_t=\alpha_1 W_{t-1}+e_t$ where $p=1$. In otherwords, when AR(p=0) the autoregression form becomes

$$
\begin{aligned}
Y_t &= \alpha_0 e_t^2+\alpha_1 Y_{t-1}e_t^2\\
Y_0 &= \alpha_0 e_0^2+\alpha_1 Y_{-1}e_0\\
E[Y_0] &= \alpha_0 E[e_0^2]+\alpha_1 E[Y_{-1}]E[e_0^2]
\end{aligned}
$$

Since the expectation values go to one, but the bounded geometric form won't be realized, meaning that we end up with

$$
\begin{aligned}
&Y_0=\alpha_0+\alpha_1 Y_{-1}\\
\text{and}\quad &Y_0\neq Y_{-1}
\end{aligned}
$$

or is not stationary and nor is $p=0$ a solution for AR(0) process of $W_t=\alpha_1 W_{t-1}+e_t$. But, for $t\geq 1$ the expectation values have stationarity where we define the autocorrelation function as

$$
\begin{aligned}
\rho(t) &= W_t/W_0\\
&\text{using recursive relation then}\\
W_t &= \alpha_1 W_{t-1}\\
\rho(t) &= \alpha_1 W_{t-1}/W_0\\
&= \alpha_1 W_{t-1}
\end{aligned}
$$

since $\rho(t=0)=1$, $\rho(1)=|\alpha_1|$, $\rho(2)=\alpha_1^2,\dots$. Since we have a converging geometric series expance of $\alpha_1^{|j|}$ and the negative lags $j=-1,\dots,-p$ converge absolutely. Generally

$$
\rho(t)_j=\begin{cases}\alpha_i^j & j\geq 0\\ \alpha_i^{|j|} & j<0\end{cases}
$$

□

7.3 Suppose that $\{Z_t\}$ is a causal stationary GARCH(p, q) process $Z_t=\sqrt{h_t}e_t$, where $\{e_t\}\sim$ IID(0, 1), $\sum_{i=1}^{p}a_i+\sum_{i=1}^{q}B_i<1$ and

$$
h_t=\alpha_0+\alpha_1 Z_{t-1}^2+\dots+\alpha_p Z_{t-p}^2+\beta_1 h_{t-1}+\dots+\beta_q h_{t-q}.
$$

a. Show that $E(Z_t^2|Z_{t-1}^2,Z_{t-2}^2,\dots)=h_t$.

b. Show that squared process $\{Z_t^2\}$ is an ARMA(m, p) process satisfying the equations

$$
\begin{aligned}
Z_t^2 = \alpha_0+(\alpha_1+\beta_1)Z_{t-1}^2+\dots+(\alpha_m+\beta_m)Z_{t-m}^2\\
+U_t-\beta_1 U_{t-1}-\dots-\beta_q U_{t-q},
\end{aligned}
$$

where $m=\max\{p,q\}$, $\alpha_j=0$ for $j>p$, $\beta_j=0$ for $j>q$, and $U_t=Z_t^2-h_t$ is white noise if $EZ_t^4<\infty$.

c. For $p\geq 1$, show that the conditional variance process $\{h_t\}$ is an ARMA(m, $p-1$) process satisfying the equations

$$
\begin{aligned}
h_t = \alpha_0+(\alpha_1+\beta_1)h_{t-1}+\dots+(\alpha_m+\beta_m)h_{t-m}+\\
V_t+\alpha_1^* V_{t-1}+\dots+\alpha_p^* V_{t-p-1},
\end{aligned}
$$

where $V_t=\alpha_1^{-1}U_{t-1}$ and $\alpha_j^*=\alpha_{j+1}/\alpha_1$ for $j=1,\dots,p-1$.

7.3 Sol^{n} For b. The ARMA(m, p) process is

$$
X_t=\sum_{i=1}^{m}\phi_i X_{t-\text{i}}+Z_t+\sum_{j=1}^{m}\theta_j Z_{j-1}.
$$

The squared process $\{Z_t^2\}$ is an ARMA(m, p)

$$
\begin{aligned}
Z_t^2 &= \alpha_0+(\alpha_1+\beta_1)Z_{t-1}^2+\dots+(\alpha_m+\beta_m)Z_{t-m}^2\\
&+Z_t^2-h_t-\beta_1(Z_{t-1}^2-h_{t-1})-\dots-\beta_q(Z_{t-q}^2-h_{t-\text{q}})
\end{aligned}
$$

Simplifying further we have

$$
\begin{aligned}
Z_t^2-Z_t^2+h_t &= \alpha_0+\alpha_1 Z_{t-1}^2+\dots+\alpha_m Z_{t-m}^2\\
&+\beta_1 h_{t-1}+\dots+\beta_q h_{t-q}\\
h_t &= \alpha_0+\alpha_1 Z_{t-1}^2+\dots+\alpha_m Z_{t-m}^2\\
&+\beta_1 h_{t-1}+\dots+\beta_q h_{t-q}\\
&= \alpha_0+\sum_{i=1}^{m}\alpha_i Z_{t-i}^2+\sum_{j=1}^{q}\beta_j h_{t-j}
\end{aligned}
$$

The $\beta_i Z_{t-i's}^2$ subtract out. This shows the general ARMA(m, p) process, $m = \max\{p, q\}$

$$\boxed{Z_t^2 - \alpha_0 + \sum_{i=1}^{m} \alpha_i Z_{t-i}^2 = Z_t^2 - h_t + \sum_{j=1}^{m} \beta_j h_{t-j}}$$

□

Part c. Sol^{n} The equation simplifies to

$$h_t = \alpha_0 + (\alpha_1 - \beta_1)h_{t-1} + \cdots + (\alpha_m + \beta_m)h_{t-m} +$$
$$Z_{t-1}^2 - h_{t-1} + \sum_{j=1}^{p-1} \alpha_{j+1}(Z_{t-p-1}^2 - h_{t-p-1})$$

$\alpha_i h_{t-i},\ i \geq 2, \ldots, p-1$ subtract out. Leaving

$$h_t = \alpha_0 + (\alpha_1 + \beta_1)h_{t-1} + \cdots + \beta_m h_{t-m} +$$
$$Z_{t-1}^2 - h_{t-1} + \sum_{j=1}^{p-1} \alpha_{j+1} Z_{t-p-1}^2.$$

At $p = 1, 2$ we derive $\alpha_{1,\,2}$ coefficients for the conditional variances and stationary processes h_t, h_{t-1}, Z_t^2, Z_{t-1}^2 and the $\alpha_i h_{t-m}$'s and $\alpha_i h_{t-p-1}$'s $i \geq 1$ will subtract out, for $m = \max\{p, q\}$ values in our ARMA(m, $p-1$) model. Therefore,

$$h_t + h_{t-1} - \sum_{i=1}^{p-1=m} \beta_i h_{t-i} = Z_{t-1}^2 + \sum_{j=1}^{p-1} \alpha_{j+1} Z_{t-p-1}^2$$

$$h_t = -h_{t-1} + \sum_{i=1}^{m} \beta_i h_{t-1} + Z_{t-1}^2 + \sum_{j=1}^{p-1} \alpha_{j+1} Z_{t-p-1}^2$$

□

Part c. 2nd Sol^{n} The GARCH(p, q) process is the first order of business to establish to include the sum

$$h_t = \alpha_0 + \sum_{j=1}^{m} (\alpha_i + \beta_i) h_{t-\text{i}}$$

we observe since $U_t = Z_t^2 - h_t$ then we can let $Z_{t-i}^2 = U_{t-i} + h_{t-i}$. But we see now that the substitution of $V_t = \alpha_1^{-1} U_{t-1}$ by the problems's instructions is derived by this very algebra that is considered. The GARCH(p, q) process

$$h_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Z_{t-i}^2 + \sum_{j=1}^{q} \beta_j h_{t-j}$$

is now

$$h_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i (U_{t-i} + h_{t-i}) + \sum_{j=1}^{q} \beta_j h_{t-\text{i}}$$

and group like terms

$$h_t = \alpha_0 + \sum_{i=1}^{\max\{p,\,q\}} (\alpha_i + \beta_i) h_{t-i} + \sum_{j=1}^{q} \alpha_j U_{t-i}$$

Letting $V_t = \alpha_1^{-1} U_{t-1}$ means $V_t = \alpha_1^{-1}(Z_{t-1}^2 - h_{t-1})$ and letting $\alpha_j^* = \alpha_{j+1}/\alpha_1 \quad j = 1, \ldots, p-1$ has the series representation

$$h_t = \alpha_0 + \sum_{i=1}^{\max\{p,\,q\}} (\alpha_i + \beta_i) h_{t-i} + V_t + \sum_{j=1}^{p-1} \alpha_j^* V_{t-j}$$

which represents

$$h_t = \alpha_0 + \sum_{i=1}^{\max\{p,\,q\}} (\alpha_i + \beta_i) h_{t-i} + Z_{t-1}^2 - h_{t-1} +$$
$$\alpha_2(Z_{t-2}^2 - h_{t-2}) + \cdots + \alpha_p(Z_{t-p-1}^2 - h_{t-p-1})$$

where $\alpha_{j+1} U_{t-1-\text{j}} = \alpha_1 \alpha_j^* V_{t-j}$, $j = 1, \ldots, p-1$. Thus, these set of equations satisfy

$$h_t = \alpha_0 + (\alpha_1 + \beta_1) h_{t-i} + \cdots + (\alpha_m + \beta_m) h_{t-m}$$
$$V_t + \alpha_1^* V_{t-1} + \cdots + \alpha_{p-1}^* V_{t-p-1}.$$

□

7.4 To each of the seven components of the multivariate time series file as STOCK7.TSM, fit an ARMA model driven by GARCH noise. Compare the fitted models for different series and provide a comment on the differences. (For exporting components of a multivariate timeseries to a univariate project, see the topic *Getting started* in the PDF file ITSM_HELP which is included in the ITSM software package.

7.4 Sol^{n}

AAPL, MSFT, and NVDA stock raw timeseries Closing data for dates *2020-01-01* to *2024-12-24* and the daily returns, calculated as `1 - n-1/n`, for `n` days, and the cumulative daily returns calculated as $\sum_{i=1}^{n} \texttt{daily_returns}_i$, respectively. A note for when I set the orders $\{p = 2, q = 2\}$ of the GARCH driven ARMA model for MSFT the α and β coefficients and persistence parameters gave $\sum \alpha, \beta = 0.1 + 0.35 = 0.45$ for the GARCH volatility portion of the model; much less than 1, indicative of less volatility and is an unrealistic representation to model to accurately predict market behavior. In other words, persistent, clustering behavior of real financial market volatility, where $\sum \alpha, \beta < 1$ or $\sum \alpha, \beta \approx 1$, close to 1 for stationarity.

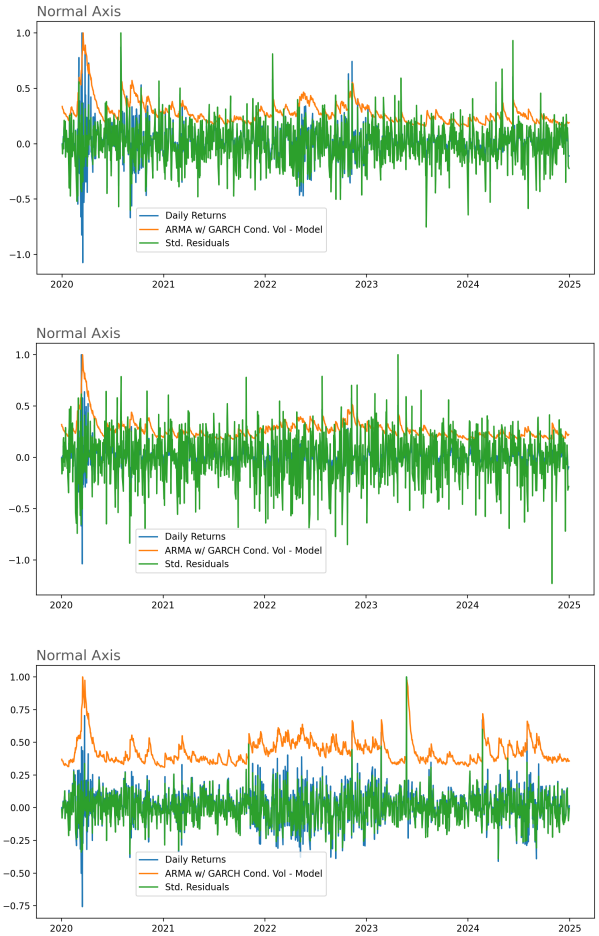

Figure 1: Visual inspection of *Daily Returns* with the fitted ARMA model driven by GARCH noise and the *Cumulative Daily Returns* with their axes normalized.

Below are tabulated comparisons of the statistical processes and findings of the individual market tickers AAPL, MSFT, and NVDA.

```
--- ARMA Model Summary ---
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 1257
Model:                 ARIMA(1, 0, 1)   Log Likelihood                3145.233
Date:                Tue, 30 Dec 2025   AIC                          -6282.466
Time:                        14:26:57   BIC                          -6261.920
Sample:                             0   HQIC                         -6274.744
                               - 1257
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0012      0.001      2.298      0.022       0.000       0.002
ar.L1         -0.1469      0.155     -0.950      0.342      -0.450       0.156
ma.L1          0.0346      0.160      0.216      0.829      -0.279       0.348
sigma2         0.0004   9.29e-06     42.245      0.000       0.000       0.000
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):              1081.11
Prob(Q):                              1.00   Prob(JB):                         0.00
Heteroskedasticity (H):               0.30   Skew:                             0.08
Prob(H) (two-sided):                  0.00   Kurtosis:                         7.54
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 2: *Daily Returns* and *Exchange Rates* for AAPL

```
--- GARCH Noise Summary on Residuals ---
                     Zero Mean - GARCH Model Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.000
Mean Model:                 Zero Mean   Adj. R-squared:                  0.001
Vol Model:                      GARCH   Log-Likelihood:                3272.86
Distribution:                  Normal   AIC:                          -6539.72
Method:            Maximum Likelihood   BIC:                          -6524.31
                                        No. Observations:                 1257
Date:                Tue, Dec 30 2025   Df Residuals:                     1257
Time:                        14:26:57   Df Model:                            0
                          Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega      7.8558e-06  5.375e-11  1.461e+05      0.000 [7.856e-06,7.856e-06]
alpha[1]       0.1000  2.953e-02      3.386  7.094e-04   [4.211e-02,  0.158]
beta[1]        0.8800  2.575e-02     34.176 5.459e-256     [  0.830,  0.930]
============================================================================

Covariance estimator: robust
/Applications/anaconda3/lib/python3.13/site-packages/arch/univariate/base.py:694
poorly scaled, which may affect convergence of the optimizer when
estimating the model parameters. The scale of y is 0.0003928. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 100 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
```

Figure 3: *Daily Returns* and *Exchange Rates* for AAPL. There is employed a simple `GARCH(1, 1)` volatility process. We observe a persistence $\sum \alpha, \beta = 0.1 + 0.88$ close to 1, indicative of high persistence i.e. volatility shocks lasting long.

```
--- ARMA Model Summary ---
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 1257
Model:                 ARIMA(1, 0, 1)   Log Likelihood                3014.463
Date:                Tue, 30 Dec 2025   AIC                          -6020.927
Time:                        14:27:05   BIC                          -6000.381
Sample:                             0   HQIC                         -6013.205
                               - 1257
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0013      0.001      2.328      0.020       0.000       0.002
ar.L1         -0.1751      0.150     -1.170      0.242      -0.468       0.118
ma.L1          0.0617      0.155      0.398      0.691      -0.242       0.365
sigma2         0.0005   1.12e-05     43.265      0.000       0.000       0.001
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):              1163.07
Prob(Q):                              1.00   Prob(JB):                         0.00
```

```
Heteroskedasticity (H):               0.27   Skew:                             0.07
Prob(H) (two-sided):                  0.00   Kurtosis:                         7.71
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 4: *Daily Returns* and *Exchange Rates* for AAPL

```
--- GARCH Noise Summary on Residuals ---
                     Zero Mean - GARCH Model Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.000
Mean Model:                 Zero Mean   Adj. R-squared:                  0.001
Vol Model:                      GARCH   Log-Likelihood:                3145.91
Distribution:                  Normal   AIC:                          -6285.82
Method:            Maximum Likelihood   BIC:                          -6270.41
                                        No. Observations:                 1257
Date:                Tue, Dec 30 2025   Df Residuals:                     1257
Time:                        14:27:05   Df Model:                            0
                          Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega      1.1664e-05  7.959e-12  1.465e+06      0.000 [1.166e-05,1.166e-05]
alpha[1]       0.0811  1.962e-02      4.134  3.570e-05   [4.265e-02,  0.120]
beta[1]        0.8937  1.664e-02     53.712      0.000     [  0.861,  0.926]
============================================================================

Covariance estimator: robust
/Applications/anaconda3/lib/python3.13/site-packages/arch/univariate/base.py:694
poorly scaled, which may affect convergence of the optimizer when
estimating the model parameters. The scale of y is 0.0004836. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 100 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
Out[766]: Text(0.075, 1, 'Normal Axis')
```

Figure 5: *Daily Returns* and *Exchange Rates* for AAPL

```
--- ARMA Model Summary ---
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 1257
Model:                 ARIMA(1, 0, 1)   Log Likelihood                3205.935
Date:                Mon, 29 Dec 2025   AIC                          -6403.871
Time:                        22:23:35   BIC                          -6383.325
Sample:                             0   HQIC                         -6396.149
                               - 1257
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0010      0.000      2.171      0.030    9.64e-05       0.002
ar.L1         -0.1285      0.090     -1.427      0.154      -0.305       0.048
ma.L1         -0.0543      0.096     -0.565      0.572      -0.243       0.134
sigma2         0.0004   8.85e-06     40.270      0.000       0.000       0.000
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):              1246.87
Prob(Q):                              1.00   Prob(JB):                         0.00
Heteroskedasticity (H):               0.36   Skew:                            -0.10
Prob(H) (two-sided):                  0.00   Kurtosis:                         7.88
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 6: *Daily Returns* and *Exchange Rates* for MSFT

```
--- GARCH Noise Summary on Residuals ---
                       Zero Mean - GARCH Model Results
==================================================================================
Dep. Variable:                      y   R-squared:                       0.000
Mean Model:                 Zero Mean   Adj. R-squared:                  0.001
Vol Model:                      GARCH   Log-Likelihood:                3337.38
Distribution:                  Normal   AIC:                          -6668.76
Method:            Maximum Likelihood   BIC:                          -6653.35
                                        No. Observations:                 1257
Date:                Mon, Dec 29 2025   Df Residuals:                     1257
Time:                        22:23:35   Df Model:                            0
                              Volatility Model
==================================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------------
omega      1.2862e-05  5.024e-12  2.560e+06      0.000 [1.286e-05,1.286e-05]
alpha[1]       0.0936  2.157e-02      4.337  1.445e-05   [5.128e-02,  0.136]
beta[1]        0.8673  1.961e-02     44.221      0.000     [  0.829,  0.906]
==================================================================================

Covariance estimator: robust
/Applications/anaconda3/lib/python3.13/site-packages/arch/univariate/base.py:694
poorly scaled, which may affect convergence of the optimizer when
estimating the model parameters. The scale of y is 0.0003566. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 100 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
```

Figure 7: *Daily Returns* and *Exchange Rates* for MSFT. There is employed a simple `GARCH(1, 1)` volatility process. We observe a persistence $\sum \alpha, \beta \approx 0.09 + 0.86$ close to 1, indicative of high persistence i.e. volatility shocks lasting long.

```
--- ARMA Model Summary ---
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 1257
Model:                 ARIMA(1, 0, 1)   Log Likelihood                3086.602
Date:                Mon, 29 Dec 2025   AIC                          -6165.204
Time:                        22:23:42   BIC                          -6144.658
Sample:                             0   HQIC                         -6157.482
                               - 1257
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0011      0.001      2.223      0.026       0.000       0.002
ar.L1         -0.1581      0.085     -1.851      0.064      -0.326       0.009
ma.L1         -0.0316      0.092     -0.344      0.731      -0.211       0.148
sigma2         0.0004   1.07e-05     40.266      0.000       0.000       0.000
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):              1259.54
Prob(Q):                              1.00   Prob(JB):                         0.00
Heteroskedasticity (H):               0.32   Skew:                            -0.11
Prob(H) (two-sided):                  0.00   Kurtosis:                         7.90
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 8: *Daily Returns* and *Exchange Rates* for MSFT

```
--- GARCH Noise Summary on Residuals ---
                       Zero Mean - GARCH Model Results
==================================================================================
Dep. Variable:                      y   R-squared:                       0.000
Mean Model:                 Zero Mean   Adj. R-squared:                  0.001
Vol Model:                      GARCH   Log-Likelihood:                3206.29
Distribution:                  Normal   AIC:                          -6406.59
Method:            Maximum Likelihood   BIC:                          -6391.18
                                        No. Observations:                 1257
Date:                Mon, Dec 29 2025   Df Residuals:                     1257
Time:                        22:23:42   Df Model:                            0
                              Volatility Model
==================================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------------
omega      8.6238e-06  1.576e-11  5.471e+05      0.000 [8.624e-06,8.624e-06]
alpha[1]       0.1000  3.468e-02      2.883  3.936e-03   [3.202e-02,  0.168]
beta[1]        0.8800  3.130e-02     28.112 7.009e-174     [  0.819,  0.941]
==================================================================================

Covariance estimator: robust
/Applications/anaconda3/lib/python3.13/site-packages/arch/univariate/base.py:694
poorly scaled, which may affect convergence of the optimizer when
estimating the model parameters. The scale of y is 0.0004312. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 100 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
Out[762]: Text(0.075, 1, 'Normal Axis')
```

Figure 9: *Daily Returns* and *Exchange Rates* for MSFT

```
--- ARMA Model Summary ---
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 1257
Model:                 ARIMA(1, 0, 1)   Log Likelihood                2475.896
Date:                Tue, 30 Dec 2025   AIC                          -4943.792
Time:                        14:34:47   BIC                          -4923.246
Sample:                             0   HQIC                         -4936.070
                               - 1257
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0031      0.001      3.336      0.001       0.001       0.005
ar.L1         -0.5131      0.186     -2.758      0.006      -0.878      -0.148
ma.L1          0.4242      0.197      2.154      0.031       0.038       0.810
sigma2         0.0011   2.85e-05     39.955      0.000       0.001       0.001
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):               828.05
Prob(Q):                              0.98   Prob(JB):                         0.00
Heteroskedasticity (H):               0.99   Skew:                             0.42
Prob(H) (two-sided):                  0.90   Kurtosis:                         6.89
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 10: *Daily Returns* and *Exchange Rates* for NVDA

```
--- GARCH Noise Summary on Residuals ---
                       Zero Mean - GARCH Model Results
==================================================================================
Dep. Variable:                      y   R-squared:                       0.000
Mean Model:                 Zero Mean   Adj. R-squared:                  0.001
Vol Model:                      GARCH   Log-Likelihood:                2521.30
Distribution:                  Normal   AIC:                          -5036.60
Method:            Maximum Likelihood   BIC:                          -5021.19
                                        No. Observations:                 1257
Date:                Tue, Dec 30 2025   Df Residuals:                     1257
Time:                        14:34:47   Df Model:                            0
                              Volatility Model
==================================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------------
omega      7.6826e-05  2.023e-05      3.797  1.465e-04 [3.717e-05,1.165e-04]
alpha[1]       0.0874  2.447e-02      3.573  3.533e-04   [3.947e-02,  0.135]
beta[1]        0.8475  2.083e-02     40.687      0.000     [  0.807,  0.888]
==================================================================================

Covariance estimator: robust
/Applications/anaconda3/lib/python3.13/site-packages/arch/univariate/base.py:694
poorly scaled, which may affect convergence of the optimizer when
estimating the model parameters. The scale of y is 0.001139. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 10 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
```

Figure 11: *Daily Returns* and *Exchange Rates* for NVDA. There is employed a simple `GARCH(1, 1)` volatility process. We observe a persistence $\sum \alpha, \beta \approx 0.09 + 0.8$ close to 1, indicative of high persistence i.e. volatility shocks lasting long.

```
--- ARMA Model Summary ---
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 1257
Model:                 ARIMA(1, 0, 1)   Log Likelihood                2358.506
Date:                Tue, 30 Dec 2025   AIC                          -4709.012
Time:                        14:34:52   BIC                          -4688.466
Sample:                             0   HQIC                         -4701.290
                               - 1257
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0034      0.001      3.385      0.001       0.001       0.005
ar.L1         -0.4859      0.190     -2.559      0.011      -0.858      -0.114
ma.L1          0.3934      0.201      1.961      0.050       0.000       0.787
sigma2         0.0014   3.49e-05     39.317      0.000       0.001       0.001
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):               718.96
Prob(Q):                              0.98   Prob(JB):                         0.00


Heteroskedasticity (H):               0.88   Skew:                             0.37
Prob(H) (two-sided):                  0.21   Kurtosis:                         6.63
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 12: *Daily Returns* and *Exchange Rates* for NVDA

```
--- GARCH Noise Summary on Residuals ---
                       Zero Mean - GARCH Model Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.000
Mean Model:                 Zero Mean   Adj. R-squared:                  0.001
Vol Model:                      GARCH   Log-Likelihood:                2399.18
Distribution:                  Normal   AIC:                          -4792.35
Method:            Maximum Likelihood   BIC:                          -4776.94
                                        No. Observations:                 1257
Date:                Tue, Dec 30 2025   Df Residuals:                     1257
Time:                        14:34:52   Df Model:                            0
                              Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega      1.1072e-04  3.145e-05      3.521  4.301e-04 [4.908e-05,1.723e-04]
alpha[1]       0.0846  2.399e-02      3.527  4.199e-04   [3.760e-02,  0.132]
beta[1]        0.8361  2.407e-02     34.743 1.752e-264     [  0.789,  0.883]
============================================================================

Covariance estimator: robust
/Applications/anaconda3/lib/python3.13/site-packages/arch/univariate/base.py:694
poorly scaled, which may affect convergence of the optimizer when
estimating the model parameters. The scale of y is 0.001373. Parameter
estimation work better when this value is between 1 and 1000. The recommended
rescaling is 10 * y.

This warning can be disabled by either rescaling y before initializing the
model or by setting rescale=False.

  self._check_scale(resids)
Out[772]: Text(0.075, 1, 'Normal Axis')
```

Figure 13: *Daily Returns* and *Exchange Rates* for NVDA

□

7.5 Verify equation (7.3.7).

7.5 Sol^{n} To motivate the definition, we recall that the GARCH(p, q) process is the causal stationary solution of the equations

$$Z_t = \sqrt{h_t} e_t, \quad h_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Z_{t-i}^2 + \sum_{i=1}^{q} \beta_i h_{t-i},$$

where $\alpha_0 > 0$, $\alpha_1, \ldots, \alpha_p \geq 0$ and $\beta_1, \ldots, \beta_q \geq 0$. It follows that

$$(1 - \alpha(B) - \beta(B)) Z_t^2 = \alpha_0 + (1 - \beta(B)) W_t,$$

where $\{W_t := Z_t^2 - h_t\}$ is white noise, $\alpha(B) = \sum_{i=1}^{p} \alpha_i B^i$ and $\beta(B) = \sum_{i=1}^{q} \beta_i B^i$. Letting $W_t = Z_t^2 - h_t$ we see

$$(1 - \alpha(B) - \beta(B)) Z_t^2 = \alpha_0 + (1 - \beta(B)) Z_t^2 - h_t$$

$$Z_t^2 - \alpha(B) Z_t^2 - \beta(B) Z_t^2 = \alpha_0 + Z_t^2 - h_t - \beta(B) Z_t^2$$

$$+ \beta(B) h_t$$

subtract Z_t^2 from both sides and add $\beta(B) Z_t^2$ from both sides and rearrange terms, then

$$h_t = \alpha_0 + \alpha(B) Z_t^2 + \beta(B) h_t$$

$$h_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i B^i Z_{t-i}^2 + \sum_{i=1}^{q} \beta_i B^i h_{t-i}$$

□

7.6 Show that the return, $Z_\Delta(t) := \log P(t) - \log P(t - \Delta)$, approximates the fractional gain, $F_\Delta(t) := (P(t) - P(t - \Delta))/P(t - \Delta)$, in this sense that

$$\frac{Z_\Delta(t)}{F_\Delta(t)} \to 1 \quad \text{as} \quad F_\Delta(t) \to 0.$$

7.6 Sol^{n} By the laws of logarithms the logarithimic return expression becomes $Z_\Delta(t) := \log P(t) - \log P(t - \Delta)$ to $Z_\Delta(t) := \log\left(\frac{P(t)}{P(t-\Delta)}\right)$, which can be rewritten

$$Z_\Delta(t) := \log\left(\frac{P(t)}{P(t - \Delta)}\right) = \log(1 + F_\Delta(t)),$$

where the *rhs* form is derived from the approximate first order Taylor series, where $F_\Delta(t) := (P(t) - P(t - \Delta))/P(t - \Delta)$. Since $\log(1) = 0.0$, which is just an additive idenity and

$$F_\Delta(t) := \frac{P(t) - P(t - \Delta)}{P(t - \Delta)} = \frac{P(t)}{P(t - \Delta)} - 1.$$

Therefore, is derived of the form given originally

$$Z_\Delta(t) := \log(1 + F_\Delta(t)) = \log\left(1 + \frac{P(t)}{P(t - \Delta)} - 1\right)$$

the 1s subtract out to give

$$Z_\Delta(t) := \log\left(\frac{P(t)}{P(t - \Delta)}\right) = \log P(t) - \log P(t - \Delta).$$

Since $\log(1) = 0.0$, then for small values of $F_\Delta(t)$ where a value of 1 is most substantial, i.e. $F_\Delta(t) << 1$, to approximate $\approx \log(1) \approx 0.0 + F_\Delta(t) = F_\Delta(t)$ for a small number $F_\Delta(t)$, or is closer to 0, and the log is more `undef` giving way to $F_\Delta(t)$. Hence, the resultant expression is

$$\frac{Z_\Delta(t)}{F_\Delta(t)} = \frac{\log\left[1 + \lim_{F_\Delta(t)\to 0}(P(t)/P(t-\Delta) - 1))\right]}{\lim_{F_\Delta(t)\to 0}(P(t)/P(t-\Delta) - 1)}$$

$$= \frac{\log(1) + F_\Delta(t)}{F_\Delta(t)}$$

$$= \frac{0.0 + F_\Delta(t)}{F_\Delta(t)} = \frac{F_\Delta(t)}{F_\Delta(t)} \longrightarrow 1$$

for increasingly small values as $F_\Delta(t)$ tends to 0. □

7.7 For the GBM model (7.5.7) with $P(0) = 1$, evaluate the mean and variance of the $P(t)$ and the mean and variance of the return, $Z_\Delta(t)$.

7.7 Sol^{n} To compute variance we need to solve the *rhs* of the following equation of the form

$$\begin{aligned}\mathrm{Var}[(c/N + \sigma\Delta_i B)^2] &= E[(c/N + \sigma\Delta_i B)^{2\cdot 2}] \\ &\quad - E[(c/N + \sigma\Delta_i B)^2]^2\end{aligned}$$

Starting with the first term on the *rhs* the steps are as follows

$$= (c/N+\sigma\Delta_i B)(c/N+\sigma\Delta_i B)(c/N+\sigma\Delta_i B)(c/N+\sigma\Delta_i B)$$

$$= \left(\frac{c^2}{N^2} + 2\frac{c}{N}\sigma\Delta_i B + \sigma^2\Delta_i^2 B^2\right)\left(\frac{c^2}{N^2} + \frac{2c\sigma\Delta_i B}{N} + \sigma^2\Delta_i^2 B^2\right)$$

$$= \frac{c^4}{N^4}\frac{c^2}{N^2}\frac{c}{N}2\sigma\Delta_i B + \frac{c^2}{N^2}\sigma^2\Delta_i^2 B^2 + \frac{c^2}{N^2}2\frac{c}{N}\sigma\Delta_i B +$$

$$4\frac{c^2}{N^2}\sigma^2\Delta_i^2 B^2 + \frac{c^2}{N^2}\sigma^2\Delta_i^2 B^2 + 2\frac{c}{N}\sigma^3\Delta_i^3 B^3 + \sigma^4\Delta_i^4 B^4$$

$$= \frac{c^4}{N^4} + \frac{4c^2}{N^2}\frac{c}{N}\sigma E[\Delta_i B] + 6\frac{c^2}{N^2}\sigma^2 E[\Delta_i^2 B^2] + \frac{2\sigma^3 c}{N}E[\Delta_i^3 B^3]$$

$$+\sigma^4 E[\Delta_i^4 B^4]$$

The last term reduces to

$$\sigma^4 E\left[\frac{1}{N^2}\sum(\Delta_i B)^4\right] = 3\sigma^3/N^2$$

where

$$\frac{1}{N}\frac{1}{N}\sum\sum(\Delta_i B)^{2\cdot 2} \implies \frac{1}{N}\sum(\Delta_i B)^2\frac{1}{N}\sum(\Delta_i B)^2$$

and then the first term on the *rhs* reduces to

$$= \frac{c^4}{N^4} + \frac{4c^2}{N^2}\frac{c}{N}\cdot 0 + 6\frac{c^2}{N^2}\sigma^2\cdot 1 + \frac{2\sigma^3 c}{N}\cdot 0 + \frac{2\sigma^4 c}{N^2}\cdot 3$$

$$= \frac{c^4}{N^4} + \frac{1}{N}\frac{6c^2}{N^2}\sigma^2 + \frac{1}{N}\sigma^4 3$$

where the middle term has its second moment calculated as

$$\frac{6c^2}{N^2}E\left[\frac{1}{N}\sum(\sigma\Delta_i B)^2\right] \implies \frac{1}{N}\frac{6c^2}{N^2}\sigma^2 E\left[\sum(\Delta_i B)^2\right]$$

For second term on the *rhs*, since we know that

$$E[(c/N + \sigma\Delta_i B)^2] = \frac{c^2}{N^2} + \frac{\sigma^2}{N}$$

and squaring gives

$$= \left(\frac{c^2}{N^2} + \frac{\sigma^2}{N}\right)\left(\frac{c^2}{N^2} + \frac{\sigma^2}{N}\right)$$

$$= \left(\frac{c^4}{N^4} + 2\frac{c^2\sigma^2}{N^3} + \frac{\sigma^4}{N^2}\right)$$

and we have for the variance

$$\mathrm{Var}[(c/N + \sigma\Delta_i B)^2] = \frac{c^4}{N^4} + \frac{1}{N}\frac{6c^2}{N^2}\sigma^2 + \frac{1}{N}\sigma^4 3 -$$

$$\left(\frac{c^4}{N^4} + 2\frac{c^2\sigma^2}{N^3} + \frac{\sigma^4}{N^2}\right)$$

$$= \frac{4c^2\sigma^2}{N^3} + \frac{3\sigma^4}{N^2} - \frac{\sigma^4}{N^2}$$

$$= \boxed{\frac{4c^2\sigma^2}{N^3} + \frac{2\sigma^4}{N^2}}$$

□

7.8 If h is any second-order stationary non-negative volatility process with mean ξ, variance ω^2 and autocorrelation function ρ, verify the relations (7.5.16)-(7.5.18).

7.8 Sol^{n}

$$EI_n = E\int_{(n-1)\Delta}^{n\Delta}\int_{-\infty}^{t} e^{\lambda(t-u)}\,L(u)$$

$$= \int_{(n-1)\Delta}^{n\Delta}\mu\int_{-\infty}^{\infty} e^{\lambda t}\,\mathrm{d}t = \mu\frac{1}{|\lambda|}$$

for $h(t) = \int_{-\infty}^{t} e^{\lambda(t-u)}\,L(u) = \int_{-\infty}^{\infty} e^{\lambda t}\,\mathrm{d}t$ where $\mu = EL(1)^2$ where we restrict to the second moment because higher moments typically don't exist in finance. Now

$$\frac{\mu}{|\lambda|}\int_{(n-1)\Delta}^{n\Delta}\mathrm{d}t = \frac{\mu}{|\lambda|}\,t\big|_{(n-1)\Delta}^{n\Delta} = \frac{\mu}{|\lambda|}\Delta = \xi\Delta$$

for $\xi = Eh(t) = \mu/|\lambda|$ □

Next we see

$$Cov\left(\int_{-\infty}^{\infty} f(t-u)\,\mathrm{d}t, \int_{-\infty}^{\infty} g(t-u)\,\mathrm{d}t\right)$$

$$= \sigma^2\int_{-\infty}^{\infty} f(u)g(u)\,\mathrm{d}u$$

Taking g as

$$g := \begin{cases} e^{\lambda x}, & x \geq 0, \\ 0 & \text{otherwise} \end{cases}, \quad \& \quad f(x) = g(x+s),\ x \in \mathbb{R}$$

therefore, the autocovariance function is

$$Cov(h(t), h(t+s)) = \sigma^2\int_{-\infty}^{\infty} g(x+s)g(x)\,\mathrm{d}x$$

$$= \sigma^2\int_{-\infty}^{\infty} e^{\lambda(t-u)}e^{\lambda(t-u+s)}\,\mathrm{d}L(u)$$

$$= \sigma^2\int_{0}^{t} e^{\lambda(t-u+t-u+s}\,\mathrm{d}L(u)$$

$$= \sigma^2\int_{0}^{t} e^{\lambda(2(t-u)+s}\,\mathrm{d}L(u)$$

$$= \sigma^2\int_{0}^{t} \frac{1}{2\lambda}e^{\lambda 2t-2\lambda u+\lambda s}\,\mathrm{d}L(u)$$

$$= \sigma^2/(2|\lambda|)\left[e^{\lambda 2t-2\lambda u+\lambda s}\right]_0^t$$

which leaves us with

$$= \frac{e^{\lambda s}(e^{2\lambda t}-1)\sigma^2}{2\lambda} = \frac{\sigma^2}{2|\lambda|}\cdot e^{\lambda s} = \frac{\sigma^2}{2|\lambda|}\rho(s), \quad s \geq 0$$

where we are only interested in the s-terms, volatility, for autocorrelation $\rho(s) = e^{\lambda s}$. Since

$$r(t) := \int_0^t\int_0^y \rho(u)\,\mathrm{d}u\mathrm{d}y$$

$$r(t) := \int_0^t\int_0^y e^{\lambda s}\,\mathrm{d}s\mathrm{d}y$$

$$r(t) := \frac{1}{\lambda^2}(e^{\lambda t}-1-\lambda t)$$

$$\gamma_I(k) = \omega^2[r((k+1)\Delta) - 2r(k\Delta) + r((k-1)\Delta)],\ \text{if } k \geq 1$$

$$= \omega^2\cdot\Bigg(\frac{1}{\lambda^2}(e^{\lambda(k+1)\Delta}-1-\lambda(k+1)\Delta)$$

$$-2\frac{1}{\lambda^2}(e^{\lambda(k\Delta)}-1-\lambda(k\Delta))$$

$$+\frac{1}{\lambda^2}(e^{\lambda(k-1)\Delta}-1-\lambda(k-1)\Delta))\Bigg)$$

Let $k = 1$, then

$$= \omega^2\cdot\Bigg(\frac{1}{\lambda^2}(e^{\lambda 2\Delta}-1-\lambda 2\Delta) - \frac{2}{\lambda^2}(e^{\lambda\Delta}-1-\lambda\Delta)$$

$$+\frac{1}{\lambda^2}(e^{\lambda 0\Delta}-1-\lambda 0\Delta)\Bigg)$$

$$= \omega^2\cdot\Bigg(\frac{1}{\lambda^2}(e^{2\lambda\Delta}-1-2\lambda\Delta) - \frac{2}{\lambda^2}(e^{\lambda\Delta}-1-\lambda\Delta)$$

$$+\frac{1}{\lambda^2}(1-1)\Bigg)$$

$$= \omega^2\cdot\left(\frac{1}{\lambda^2}(e^{2\lambda\Delta}-1-2\lambda\Delta) - \frac{2}{\lambda^2}e^{\lambda\Delta} + \frac{2}{\lambda^2} + \frac{2\lambda\Delta}{\lambda^2}\right)$$

$$= \frac{1}{\lambda^2}\omega^2\cdot(e^{\lambda\Delta}-1)(e^{\lambda\Delta}-1) = \frac{\omega^2}{\lambda^2}(e^{2\lambda\Delta}-2e^{\lambda\Delta}+1)$$

In general

$$= \frac{\sigma^2}{2|\lambda|}\cdot\frac{\sigma^2}{\lambda^2}\cdot \mathrm{e}^{(k-1)\lambda\Delta}(1-e^{\lambda\Delta})^2, \quad k \geq 1$$

$$\frac{\sigma^2}{2|\lambda|^3} \cdot e^{(k-1)\lambda\Delta}(1-e^{\lambda\Delta})^2, \quad k \geq 1$$

because

$$e^{(k-1)}(e^{2\lambda\Delta} - 2e^{\lambda\Delta} + 1)$$

$$= e^{k\lambda\Delta+\lambda\Delta} - 2e^{k\lambda\Delta} + e^{k\lambda\Delta-\lambda\Delta}.$$

□

7.9 Use (7.5.26) and (7.5.27) to evaluate the mean and autocovariance function of the stationary Ornstein-Uhlenbeck process (7.5.23).

7.9 Sol^{n} Equation (7.5.23) is

$$h(t) = \int_{\infty}^{t} e^{\lambda(t-u)}\,\mathrm{d}L(u),$$

$\lambda < 0$ and $EL(1)^2 < \infty$. We rewrite $h(t)$ as

$$h(t) = \int_{-\infty}^{\infty} g(t-u)\,\mathrm{d}L(u),$$

where

$$g(t) := \begin{cases} e^{\lambda t}, & \text{if } t \geq 0, \\ 0, & \text{otherwise} \end{cases}$$

g in the rewritten $h(t)$ equation is called a kerne function. If $EL(1)^2 < \infty$, as we will assume for now on, and if f and g are integrable and square integrable functions on $\mathbb{R}$, we have

$$E\int_{-\infty}^{\infty} f(t-u)\,\mathrm{d}L(u) = \mu\int_{-\infty}^{\infty} f(t)\,\mathrm{d}L(u)$$

and

$$Cov\left(\int_{-\infty}^{\infty} f(t-u)\,\mathrm{d}t, \int_{-\infty}^{\infty} g(t-u)\,\mathrm{d}t\right)$$

$$= \sigma^2\int_{-\infty}^{\infty} f(u)g(u)\,\mathrm{d}u$$

where $\mu = EL(u)$ & $\sigma^2 = \mathrm{Var}(L(1))$. Taking $f(t) = g(s+t)$, $t \in \mathbb{R}$, we find that the mean & autocovariance function of the volatility process $\{h(t)\}$ are $\xi = Eh(t) = \mu\int e^{\lambda(t-u)}\,\mathrm{d}u = \mu/|\lambda|$, where we restrict to the second moment, $E(h(t))^2 = 1$, since higher moments in finance typically don't exist. Next

$$Cov(h(t+s), h(t)) = \sigma^2\int e^{\lambda(t-u)}e^{\lambda(t-u+s)}\,\mathrm{d}u$$

$$= \sigma^2 \cdot \int e^{\lambda[2(t-u)+s]}\,\mathrm{d}u$$

$$= \sigma^2 \cdot \frac{1}{2|\lambda|}\left[e^{2\lambda t-2\lambda u+\lambda s}\right]_0^t$$

$$= \sigma^2 \cdot 2|\lambda|^{-1} \cdot e^{\lambda s}(e^{2\lambda t} - 1)$$

$$= \sigma^2/(2|\lambda|)e^{\lambda s} = \omega^2\rho(s),$$

$\omega^2 = \frac{\sigma^2}{2|\lambda|}$, the autocorrelation function, $\rho(s) = e^{\lambda s}$, of the Orstein-Uhlenbeck SV model. (Physical meaning: This form often appears in first-order systems, ex. the velocity of a particle undergoing Brownian motion (the Orstein-Uhlenbeck process).)

Substitute ρ into

$$r := \int_0^t \rho(u)\,\mathrm{d}u\,\mathrm{d}y.$$

to give

$$r := \frac{1}{\lambda^2}(e^{\lambda t} - 1 - \lambda t).$$

In particular we find that

$$EY_n^2 = EI_n = E\int_{(n-1)\Delta}^{n\Delta}\int_{-\infty}^{t} e^{\lambda(t-u)}\,L(u)$$

$$= \int_{(n-1)\Delta}^{n\Delta} \mu\int_{-\infty}^{\infty} e^{\lambda t}\,\mathrm{d}t = \mu\frac{1}{|\lambda|}$$

for $h(t) = \int_{-\infty}^{t} e^{\lambda(t-u)}\,L(u) = \int_{-\infty}^{\infty} e^{\lambda t}\,\mathrm{d}t$ where $\mu = EL(1)^2$ where we restrict to the second moment because higher moments typically don't exist in finance. Now

$$\frac{\mu}{|\lambda|}\int_{(n-1)\Delta}^{n\Delta}\mathrm{d}t = \frac{\mu}{|\lambda|}\,t\Big|_{(n-1)\Delta}^{n\Delta} = \frac{\mu}{|\lambda|}\Delta = \xi\Delta$$

for $\xi = Eh(t) = \mu/|\lambda|$ $\square$

$$\gamma_I(k) = \omega^2[r((k+1)\Delta) - 2r(k\Delta) + r((k-1)\Delta)], \text{ if } k \geq 1$$

$$= \omega^2 \cdot \Bigg(\frac{1}{\lambda^2}(e^{\lambda(k+1)\Delta} - 1 - \lambda(k+1)\Delta)$$

$$- 2\frac{1}{\lambda^2}(e^{\lambda(k\Delta)} - 1 - \lambda(k\Delta))$$

$$+ \frac{1}{\lambda^2}(e^{\lambda(k-1)\Delta} - 1 - \lambda(k-1)\Delta))\Bigg)$$

Let $k = 1$, then

$$= \omega^2 \cdot \Bigg(\frac{1}{\lambda^2}(e^{\lambda 2\Delta} - 1 - \lambda 2\Delta) - \frac{2}{\lambda^2}(e^{\lambda\Delta} - 1 - \lambda\Delta)$$

$$+ \frac{1}{\lambda^2}(e^{\lambda 0\Delta} - 1 - \lambda 0\Delta)\Bigg)$$

$$= \omega^2 \cdot \Bigg(\frac{1}{\lambda^2}(e^{2\lambda\Delta} - 1 - 2\lambda\Delta) - \frac{2}{\lambda^2}(e^{\lambda\Delta} - 1 - \lambda\Delta)$$

$$+ \frac{1}{\lambda^2}(1-1)\Bigg)$$

$$= \omega^2 \cdot \Bigg(\frac{1}{\lambda^2}(e^{2\lambda\Delta} - 1 - 2\lambda\Delta) - \frac{2}{\lambda^2}e^{\lambda\Delta} + \frac{2}{\lambda^2} + \frac{2\lambda\Delta}{\lambda^2}\Bigg)$$

$$= \frac{1}{\lambda^2}\omega^2 \cdot (e^{\lambda\Delta} - 1)(e^{\lambda\Delta} - 1) = \frac{\omega^2}{\lambda^2}(e^{2\lambda\Delta} - 2e^{\lambda\Delta} + 1)$$

In general

$$\gamma_{Y^2}(k) = \gamma_I(k) = \frac{\sigma^2}{2|\lambda|} \cdot \frac{\sigma^2}{\lambda^2} \cdot \mathrm{e}^{(k-1)\lambda\Delta}(1 - e^{\lambda\Delta})^2, \quad k \geq 1$$

$$\frac{\sigma^2}{2|\lambda|^3} \cdot e^{(k-1)\lambda\Delta}(1 - e^{\lambda\Delta})^2, \quad k \geq 1$$

because

$$e^{(k-1)}(e^{2\lambda\Delta} - 2e^{\lambda\Delta} + 1)$$

$$= e^{k\lambda\Delta + \lambda\Delta} - 2e^{k\lambda\Delta} + e^{k\lambda\Delta - \lambda\Delta}.$$

The validity of the latter expressions for $k \geq 1$ and not for $k = 0$ indicates that both the squared return sequence $\{Y^2\}$ and the integrated volatility sequence $\{I_n\}$ have the autocovariances fo ARMA(1, 1) processes. This demonstrates, for this particular model, the covariance structure of the sequence $\{Y^2\}$ and the consquent dependence of the white-noise sequence $\{Y_n\}$.

$\square$

7.10 If h is the stationary Ornstein-Uhlenbeck process (7.5.23) and s is any fixed value in $[0, \Delta]$, show that application of the operator $\phi(B) := (1 - e^{\lambda\Delta}B)$ to the sequence $\{h(n\Delta + s), n \in \mathbb{Z}\}$ gives

$$\phi(B)h(n\Delta + s) = W_n(s),$$

where $\{W_n(s), n \in \mathbb{Z}\}$ is iid sequence,

$$W_n(s) = \int_{(n-1)\Delta+s}^{n\Delta+s} e^{\lambda(n\Delta+s-u)}\mathrm{d}L(u).$$

Deduce that the integrated volatility sequence, $I_n = \int_{-\Delta}^{0} h(n\Delta + s)\mathrm{d}s$, satisfies

$$(1 - e^{\lambda\Delta}B)I_n = \int_{-\Delta}^{0} W_n(s)\mathrm{d}s.$$

Since the *rhs* is 1-correlated, it follows from Proposition 2.1.1 that it is an MA(1) process and hence that the integrated volatility sequence is an ARMA(1, 1) process.

7.10 Sol^{n} The operator $\phi(B) := (1 - e^t B)$ of the Ornstein-Uhlenbeck process has the recursive property for the Ornstein-Uhlenbeck random volatility $h(t)$ with solution $h(t) = e^{\lambda t}h(t_0) + \int_{t_0}^{t} e^{\lambda(t-u)}L(u)$ for $t > t_0$ & satisfies differential equation $\mathrm{d}h(t) = \lambda h(t)\mathrm{d}t + \mathrm{d}L(t)$. When applied, in general

$$\phi(B)h(t) = (1 - e^t B)h(t),$$

where B is the backshift operator *s.t.* $Bh(t) = h(t_0)$. Therefore,

$$= e^{\lambda t}h(t_0) + \int_{t_0}^{t} e^{\lambda(t-u)}\, L(u)$$

$$= e^{\lambda t}h(t_0) + \int_{t_0}^{t} e^{\lambda(t-u)}\, L(u) - e^{\lambda t}h(t_0).$$

Applying the lower & upper bounds $(n-1)\Delta + s$ & $n\Delta + s$, respectively, produces the result

$$\phi(B)h(n\Delta + s) = (1 - e^{\lambda\Delta}B)h(n\Delta + s)$$

since $t-t_0 = n\Delta+s-[(n-1)\Delta+s] = n\Delta+s-n\Delta+\Delta-s = \Delta$. Then,

$$h(n\Delta+s)-e^{\lambda\Delta}Bh(n\Delta+s) \Rightarrow h(n\Delta+s)-e^{\lambda\Delta}h((n-1)\Delta+s,$$

which is rewritten

$$e^{\lambda\Delta}h((n-1)\Delta+s) + \int_{(n-1)\Delta+s}^{n\Delta+s} e^{\lambda(n\Delta+s-u}\, L(u)$$

$$- e^{\lambda\Delta}h((n-1)\Delta+s).$$

Thus, we show that

$$\phi(B)h((n-1)\Delta+s) = W_n(s)$$

where the resursive terms subtract out & $W_n(s) \sim$ iid, $s.t.\, W_n(s) = \int_{(n-1)\Delta+s}^{n\Delta+s} e^{\lambda(n\Delta+s-u}\, L(u)$ equating both sides □

7.11 For the stochastic volatility model (7.5.13) with $m = b = 0$ and second-order stationary volatility process h independent of W, establish (7.5.20) and (7.5.21).

7.11 Sol^{n} Equation 7.5.13 is

$$\mathrm{d}X(t) = [m + bh(t)]\mathrm{d}t + \sqrt{h(t)}\mathrm{d}B(t),\ t \geq 0,\ X(0) = 0$$

Quantitites associated with the model (7.5.13) which are of particular interest are the returns over time intervals of length $\Delta > 0$, i.e.

$$Y_n := X(n\Delta) - X((n-1)\Delta), \quad n \in \mathbb{N}$$

$$\sqrt{h(t)}\mathrm{d}B(t) = \mathrm{d}X(t)$$

when $m = b = 0$. The GBM model for $\{P(t), t \geq 0\}$ is defined by the Itô stochastic differential equation,

$$\mathrm{d}P(t) = P(t)[\mu\mathrm{d}t + \sigma\mathrm{d}B(t)],\ \text{with}\, P(0) > 0.$$

The solution satisfies

$$P(t) = P(0)\exp[(\mu + \sigma/2)t + \sigma B(t)],$$

it follows the log asset price $X(t) = \log P(t)$ satisfies

$$\Rightarrow \mathrm{d}X(t) = (\mu + \sigma/2)\mathrm{d}t + \sigma\mathrm{d}B(t)$$

The return for the time interval is $((n-1)\Delta, n\Delta)$

$$Y_n := X(n\Delta) - X((n-1)\Delta), \quad n \in \mathbb{N}$$

$$EY_n^2 = \int_{(n-1)\Delta+s}^{n\Delta+s} e^{\lambda(n\Delta+s-u)}\,\mathrm{d}L(u) = h(n\Delta)(\Delta_i B)^2$$

$\Delta h\, E[B^2] \longrightarrow \Delta h \cdot 1$

since $EI_n = EY_n^2$

$$I_n = \int_{(n-1)\Delta+s}^{n\Delta+s}\int_{-\infty}^{t} h(t)\,\mathrm{d}t,\ n \in \mathbb{N}$$

$$= \int_{(n-1)\Delta+s}^{n\Delta+s}\int_{-\infty}^{t} e^{\lambda(t-u)}\,\mathrm{d}L(u)\mathrm{d}t\, n \in \mathbb{N}$$

$$= \xi \int_{(n-1)\Delta+s}^{n\Delta+s} \mathrm{d}L(u)$$

$$= \xi\ [\ t\ ]_{(n-1)\Delta+s}^{n\Delta+s}$$

$$= \xi\,[n\Delta + s - n\Delta + \Delta - s]$$

$$= \xi\Delta$$

where $\xi = \mu/|\lambda|$. And

$$Cov(h(n\Delta+s), h(n\Delta)) \Rightarrow \sigma^2 \int_{(n-1)\Delta+s}^{n\Delta+s} e^{\lambda(n\Delta+s-u)}\, e^{\lambda(n\Delta-u)}\,\mathrm{d}u$$

$$= \sigma^2 \frac{1}{2|\lambda|}(e^{\lambda n\Delta+\lambda s-\lambda n\Delta} - e^{\lambda n\Delta-\lambda n\Delta-\lambda\Delta})$$

$$= \sigma^2 \frac{1}{2|\lambda|}(e^{\lambda s} - e^{\lambda\Delta})$$

$$\Rightarrow \frac{\sigma^2}{2|\lambda|}e^{\lambda s} = \omega^2\rho(s)\,, s \geq 0$$

where $\omega^2 = \sigma^2/(\,2|\lambda|)$, $\rho(s) = e^{\lambda s}$.

The autocovariance function

$$\gamma_{Y^2}(k) = \begin{cases} \omega^2[6r(\Delta) + 2\Delta^2\xi^2/\omega^2],\ \text{if}\ k = 0, \\ \omega^2[r((k+1)\Delta) - 2r(k\Delta) + r((k-1)\Delta)],\ \text{if}\ k \geq 1 \end{cases}$$

$$= \begin{cases} \omega^2[2r(\Delta) + 0.\overline{66}\Delta^2\xi^2/\omega^2], & \text{if } k = 0, \\ \omega^2[r((k+1)\Delta) - 2r(k\Delta) + r((k-1)\Delta)], & \text{if } k \geq 1 \end{cases}$$

where

$$r(t) := \int_0^t \int_0^y \rho(s)\,\mathrm{d}s\,\mathrm{d}y$$

and the Y_n^2's consequent dependence of white noise, $0.\overline{66}\Delta^2\xi^2/\omega^2$, the additive sequence $\{Y_n\}$. □

7.12 Verify that the expression (7.6.23) for $v(t,s)$ satisfies (7.6.13) and (7.6.14) and use it to write down the value of the option at time $t = 0$ and the corresponding investment strategy $\{(a_t, b_t),\ 0 \leq t \leq T\}$

7.12 Sol^{n} Substituting $\max(s-K,0)$ for $h(s)$ in the equation

$$v(t,\ s) = e^{-r(T-t)} \int h\left(se^{(r-\sigma^2/2)(T-t)+\sigma y\sqrt{T-t}}\right) \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\mathrm{d}y$$

which gives

$$v(t,\ s) = s\Phi(z_1) - K\exp(-r(T-t))\Phi(z_2)$$

where Φ is the standard normal cumulative distribution function, $\Phi(s) = \int_{-\infty}^{s} \Phi(u)\mathrm{d}u$,

$$z_1 = \frac{\log(s/K) + (r+\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \quad \text{and} \quad z_2 = z_1 - \sigma\sqrt{T-t}.$$

the steps to simplifying and solving are

$$v(t,\ s) = \int se^{(-r+r-\sigma^2/2)(T-t)+\sigma y\sqrt{T-t}} e^{-y^2/2} \frac{1}{\sqrt{2\pi}}\mathrm{d}y - Ke^{-r(T-t)} \int \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\mathrm{d}y$$

$$v(t,\ s) = \int se^{-\sigma^2/2(T-t)-\frac{1}{2}(y^2-2\sigma y\sqrt{T-t})}\mathrm{d}y/\sqrt{2\pi} - Ke^{-r(T-t)} \int \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\mathrm{d}y$$

complete the square of the exponential term: $y^2 - 2\sigma y\sqrt{T-t} = (y - \sigma\sqrt{T-t})^2 - \sigma^2(T-t)$ giving

$$v(t,\ s) = \int se^{-\sigma^2/2(T-t)-\frac{1}{2}((y-\sigma\sqrt{T-t})^2-\sigma^2(T-t))}\mathrm{d}y/\sqrt{2\pi} - Ke^{-r(T-t)} \int \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\mathrm{d}y$$

simplifying to

$$v(t,\ s) = \int se^{-\frac{1}{2}(y-\sigma\sqrt{T-t})^2} \frac{1}{\sqrt{2\pi}}\mathrm{d}y - Ke^{-r(T-t)} \int \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\mathrm{d}y$$

deriving z_1 and z_2 is done through substituting $\max(s-K,0)$ for $h(s)$, the integrand $v(t,\ s)$, and with its properties of logarithms solve for y *s.t.*

$$v(t,\ s) = \int \max\left(se^{(r-\sigma^2/2)(T-t)+\sigma y\sqrt{T-t}} - K, 0\right) \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\mathrm{d}y$$

since

$$= \max\left(\log(s)\log\left(e^{(r-\sigma^2/2)(T-t)+\sigma y\sqrt{T-t}}\right) - \log(K), 0\right)$$

$$\log(s/K) - (r-\sigma^2/2)(T+t) + \sigma y\sqrt{T-t} = 0$$

$$y = \frac{\log(s/K) - (r-\sigma^2/2)(T+t)}{\sigma\sqrt{T-t}}$$

$$y > \frac{\log(s/K) - (r-\sigma^2/2)(-T+t)}{\sigma\sqrt{T-t}} = z_2$$

and

$$z_1 = z_2 + \sigma\sqrt{T-t}$$

where

$$z_1 = \frac{\sigma^2(T-t) + \log(s/K) - (r+\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}$$

$$z_1 = \frac{\log(s/K) + (r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}$$

$$z_2 = \frac{\log(s/K) - (r-\sigma^2/2)(-T-t)}{\sigma\sqrt{T-t}}$$

we see now from equation

$$v(t,\ s) = \int se^{-\frac{1}{2}(y-\sigma\sqrt{T-t})^2} \frac{1}{\sqrt{2\pi}}\mathrm{d}y - Ke^{-r(T-t)} \int \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}}\mathrm{d}y$$

the exponentials in the first and second integrals contain z_1 and z_2, respectively, to give

$$v(t,\ s) = s\Phi(z_1) - Ke^{-r(T-t)}\Phi(z_2)$$

where $\Phi(\cdot) \sim N(0,\ 1)$. Equation 7.6.13 is

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 s^2 \frac{\partial^2 v}{\partial s^2} + rs\frac{\partial v}{\partial s} = rv.$$

The first term is calculated as

$$\frac{\partial v}{\partial t} = \frac{\partial}{\partial t} s\Phi(z_1) - Ke^{-r(T-t)}\Phi(z_2)$$

$$= s\Phi(z_1)\frac{\partial z_1}{\partial t} - \left[-rKe^{-r(T-t)}\Phi(z_2) + Ke^{-r(T-t)}\Phi(z_2)\frac{\partial z_2}{\partial t}\right]$$

$$v_t = s\Phi(z_1)\frac{\partial z_1}{\partial t} + rKe^{-r(T-t)}\Phi(z_2) - Ke^{-r(T-t)}\Phi(z_2)\frac{\partial z_2}{\partial t}$$

Using the chain rule where for instance $\Phi(z_2)\frac{\partial z_2}{\partial t}$. Note, the first term implies the calculation $s\Phi(z_1)\frac{\partial z_1}{\partial t} + 0 \cdot \Phi(z_1)$. Using the identity $s\Phi(z_1) = Ke^{-r(T-t)}\Phi(z_2)$ that hold for these specific definitions of z_1 and z_2, we find the terms $s\Phi(z_1)\partial z_1/\partial t = Ke^{-r(T-t)}\Phi(z_2)\partial z_2/\partial t$

$$\Rightarrow s\Phi(z_1)\frac{\partial z_1}{\partial t} - Ke^{-r(T-t)}\Phi(z_2)\frac{\partial z_2}{\partial t}$$

$$\Rightarrow s\Phi(z_1) - Ke^{-r(T-t)}\Phi(z_2)\left(\frac{\partial z_1}{\partial t} - \frac{\partial z_2}{\partial t}\right) \longrightarrow \frac{\sigma}{2\sqrt{T-t}} s\Phi(z_1)$$

since $z_1 - z_2 = \sigma\sqrt{T-t} \Rightarrow \partial z_1/\partial t - \partial z_2/\partial t = 1/\sigma\sqrt{T-t} \cdot 2^{-1}$. Substituting this back into the expression becomes

$$\frac{\partial v}{\partial t} = \frac{s\sigma}{2\sqrt{T-t}}\Phi(z_1) + rKe^{-r(T-t)}\Phi(z_2)$$

Finally, applying $\partial v/\partial t = -\partial v/\partial t$ (because the s & K terms are equal & have dual $\mp$ signs) yields the result

$$\frac{\partial v}{\partial t} = -\frac{s\sigma\Phi\left(\frac{\log(s/K)-(r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}\right)}{2\sqrt{T-t}} + rKe^{-r(T-t)}\Phi\left(\frac{\log(s/K)-(r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}\right)$$

Next for our second term

$$\frac{\partial^2 v}{\partial s^2} = s\Phi(z_1)\frac{\partial z}{\partial s} + \Phi(z_1)$$

$$= s\frac{\partial^2 z_1}{\partial s^2} + 0 \cdot \Phi(z_1)$$

$$= s\frac{\partial^2 z_1}{\partial s^2}$$

In addition to the second term, $1/2s^2\sigma^2\partial^2 v/\partial s^2$, we divide the 2nd PDE into two part-differentiation. First, we denote the first derivative as Delta (Δ), which simplifies significantly due to the property $s\Phi(z_1) = Ke^{-r(T-t)}\Phi(z_2)$.

Thus,

$$s\Phi(z_1) - Ke^{-r(T-t)}\Phi(z_2) = 0$$

$$= s\Phi(z_1)\frac{\partial z_1}{\partial s} + \Phi(z_1) - 0 \cdot \Phi(z_2) - Ke^{-r(T-t)}\Phi(z_2)\frac{\partial z_1}{\partial s}$$

$$= s\Phi(z_1)\frac{\partial z_1}{\partial s} + \Phi(z_1) - Ke^{-r(T-t)}\Phi(z_2)\frac{\partial z_1}{\partial s}$$

$$\Delta = \Phi(z_1) \quad \boxed{\text{since } 0 = \left(\frac{\partial z_1}{\partial s} - \frac{\partial z_2}{\partial s}\right) = \frac{1}{s\sigma\sqrt{T-t}} - \frac{1}{s\sigma\sqrt{T-t}}}$$

taking derivatives on both sides

$$\frac{\partial \Delta}{\partial s} = \Phi(z_1)\frac{\partial z_1}{\partial s}$$

$$= \Phi(z_1)\frac{\partial}{\partial s}\frac{\log(s/K) - (r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}$$

$$= \Phi(z_1)\frac{\partial}{\partial s}\frac{\log(s) - \log(K) - (r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}$$

$$= \Phi(z_1)\frac{\partial}{\partial s}\left(\frac{\log(s)}{\sigma\sqrt{T-t}} - \frac{\log(K)}{\sigma\sqrt{T-t}} - \frac{(r-\sigma^2/2)}{\sigma\sqrt{T-t}}\frac{(T-t)}{\sigma\sqrt{T-t}}\right)$$

$$= \Phi(z_1)\frac{1}{s} \cdot \frac{1}{\sigma\sqrt{T-t}} - 0 - 0$$

$$= \Phi(z_1)\frac{1}{s} \cdot \frac{1}{\sigma\sqrt{T-t}}$$

Thus, giving the result

$$\frac{\partial^2 v}{\partial s^2} = \Phi(z_1)\frac{1}{s\sqrt{T-t}} \quad \text{for} \quad \frac{1}{2}\sigma^2 s^2\frac{\partial^2 v}{\partial s^2}$$

Hence,

$$= \frac{1}{2}\sigma^2 s^2\Phi(z_1)/s\sigma\sqrt{T-t}$$

$$= \frac{\sigma s\Phi(z_1)}{2\sqrt{T-t}}.$$

Note: For $\partial s/\partial t$ at step of calculating the first term $s\Phi(z_1)\frac{\partial z_1}{\partial t} = Ke^{-r(T-t)}\frac{\partial z_2}{\partial t}$

$$s\Phi(z_1)\frac{\partial z_1}{\partial t} - Ke^{-r(T-t)}\frac{\partial z_2}{\partial t} = s\Phi(z_1) - Ke^{-r(T-t)}\left(\frac{\partial z_1}{\partial t} - \frac{\partial z_2}{\partial t}\right)$$

$$\left(\frac{\partial z_1}{\partial t} - \frac{\partial z_2}{\partial t}\right) = \frac{\partial}{\partial t}\left(\frac{\log(s/K) - (r - \sigma^2/2)(T-t) - \log(s/K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}\right)$$

$$= \frac{\partial}{\partial t}\left(\frac{\log(s/K) - \log(s/K) - rT + rt + \frac{\sigma^2}{2}T - \frac{\sigma^2}{2}t + rT - rt + \frac{\sigma^2}{2}T - \frac{\sigma^2}{2}t)}{\sigma\sqrt{T-t}}\right)$$

$$= \frac{\partial}{\partial t}\left(\frac{\log(s/K) - \log(s/K)}{\sigma\sqrt{T-t}} + \frac{\sigma^2(T-t)}{\sigma\sqrt{T-t}}\right)$$

this simplifies to

$$\frac{\partial}{\partial t}\sigma\sqrt{T-t} = -\frac{1}{2}\sigma(T-t)^{-1+1/2=-1/2} \cdot (-1)$$

$$= \frac{s\sigma\Phi(z_1)}{2\sqrt{T-t}}.$$